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29 May 2026

Recommendation Letter Diogo Poeta

Diogo Poeta joined our research team as a research assistant at the INL for the LEARN project (https://www.learnproject-heu.eu/) September 2022. Diogo primary role in our project encompassed the design of suspended dielectric nanomembranes and nanostructures for the efficient generation of third harmonic light down to the vacuum ultraviolet. Throughout his stay at INL, Diogo exhibited a great capacity in independent learning of new concepts and finite element modelling software. In this way, he managed to predict for the first time the occurrence of a special optical phenomenon referred to as anapoles in the ultraviolet wavelength domain. This resulted in a research publication in Optics Letters[1], one of the leading journals within the optics research community. In the second phase of his project, Diogo investigated the fabrication and experimental material characterization of titanium dioxide thin films of high optical quality. He linked these experimental results with the theoretical investigation of energy velocity, resulting in a manuscript currently under evaluation for journal submission.

Diogo has demonstrated himself as a motivated and engaged colleague delivering key contributions to our project aiming to develop new sensor technology for the real-time molecular selective detection of volatile organic compounds. I believe that his experience working within the international INL RTO environment equips him with a broad set of skills, both professional and interpersonal, to succeed in the next stage of his career.

Yours faithfully,



Pieter De Beule
MEng PhD DIC
Research Scientist, LEARN WP4 coordinator of the VOC sensor

[1] Poeta, Diogo, Rafael Vilarinho, Martín López-García, and Pieter A. A. De Beule. 2024. 'Optical Anapoles Excited by UV-A Illumination'. *Optics Letters* 49 (24): 7198. https://doi.org/10.1364/OL.536925.